

Guião 13 – Trabalhar com os equipamentos físicos

Objetivos:

- Trabalhar com os routers e switches do laboratório 13.
- Ligação via cabo de serie
- Configurações base
- Protocolos de routing dinâmico
- Proteção dos equipamentos com credenciais

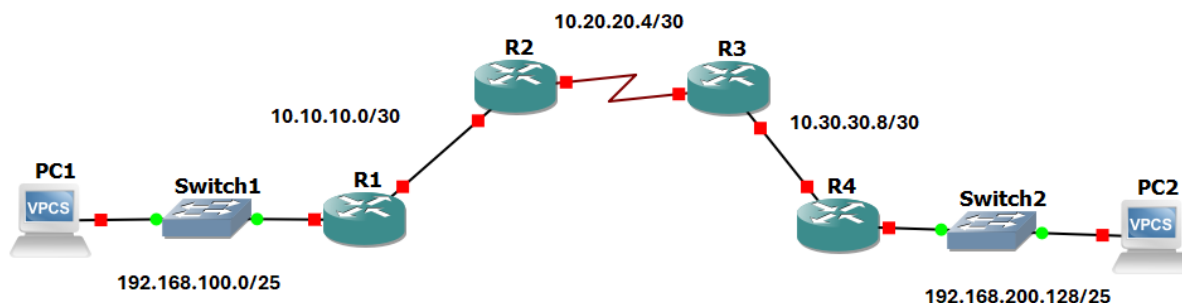


Figura 1 - diagrama de rede

1. Implemente o diagrama de rede com os equipamentos físicos da sala.
2. Configure o endereçamento de rede com as redes indicadas na figura
3. Implemente o protocolo de encaminhamento dinâmico OSPF na zona 0, para todas as redes da figura.
4. Implemente o DHCP para as redes 192.168.100.0 e 192.168.200.0 nos routers 2 e 4.
5. Implemente a autenticação para acesso à CLI e ao modo privilegiado.

1. Criar uma configuração básica no Router/Switch

- a. Atribuir um nome ao Router/Switch;

```
Switch# configure terminal  
Switch(config)# hostname Switch0
```

- b. Defina as passwords de acesso;

- i. Entre em **modo config-line**;
- ii. Defina a **password** de login para **cisco**.

```
Switch0(config)# line console 0  
Switch0(config-line)# password cisco  
Switch0(config-line)# login
```

- iii. Configure o acesso remoto vty lines 0 to 15 com a password **cisco**.

```
Switch0(config)#line vty 0 15
Switch0(config-line)# password cisco
Switch0(config-line)# login
```

2. Faça logout e tente entrar novamente na CLI

3. Gravar a configuração.

Já terminou a configuração básica do Router/Switch. Gravar a configuração no Router/Switch. Para isso deve guardar uma cópia da configuração em memória (running-config) para o ficheiro de configuração em NVRAM (startup-config). Isto garante que grava as alterações e as alterações não são perdidas quando fizer um reboot do Router/Switch.

```
Switch0#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
```

4. Repor as configurações originais.

Faça um reboot ao Switch. O switch manteve as configurações originais?

```
Switch0#reload
Switch0#show running-config
```

Executar a reposição/reinicialização das configurações do fabricante (ponto de partida) (final da aula);

```
Switch0#erase startup-config
Switch0#reload
```

6. Verifique a configuração que está a correr nos equipamentos. Verifique que está tudo correto.

7. As passwords estão visíveis... Será que existe uma forma de as ocultar? Justifique e demonstre.

EXTRA:

8. Examine as características da interface virtual VLAN1 nos switches. Acha que é possível atribuir um endereço a esta interface? Porquê? Discuta com a turma.

1.1. Examine as características da interface virtual VLAN1:

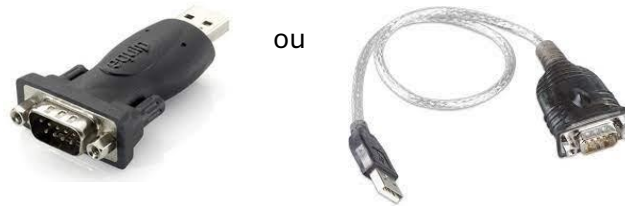
```
Switch# show interfaces vlan 1
```

NOTA: não se esquecer de repor as configurações de fábrica.

AJUDA

Para a execução dos exercícios desta ficha, os alunos deverão garantir previamente que:

- a) Todos os PC devem ter uma porta Ethernet (portaRJ45) ou um adaptador USB / Ethernet;
- b) Todos os PC tenham instalado o software Wireshark;
- c) Em cada bancada, haja 1 PC dos alunos e que um deles tenha uma porta série RS232 conforme imagem ao lado;
 - a. Em alternativa, poderão utilizar um adaptador USB/RS232:



1. Acesso aos Routers/Switch Cisco do Laboratório:

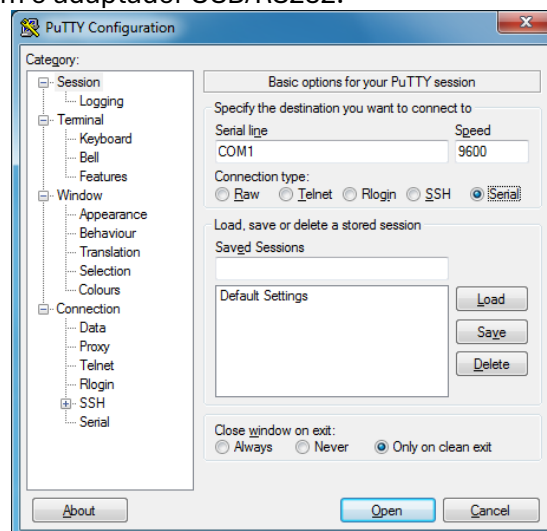
- a) O acesso aos Routers/Switch Cisco é feito, preferencialmente, através da porta de consola que está no painel traseiro do equipamento. A esta porta de consola liga-se um cabo específico, denominado “Cabo de Consola”, que por sua vez se liga aos PC pela porta série.



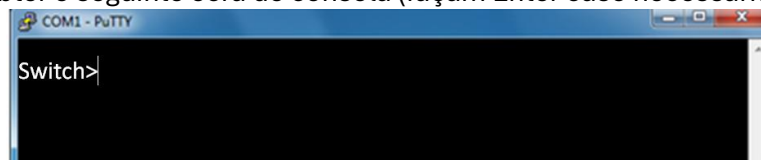
- b) Para poderem aceder ao Switch, deverão executar as seguintes ações:
- c) Ligar o cabo de consola até à porta série do PC de onde pretendem gerir o Router/Switch;



- d) Neste PC, instalar um software de acesso remoto (por exemplo o software Putty disponível em <https://www.putty.org/>). Abrir o programa e, na caixa de diálogo, selecionar a porta Serial (por exemplo a COM1 caso a porta esteja nativamente instalada no PC) ou a porta USB onde ligaram o adaptador USB/RS232:



- e) Deverão obter o seguinte ecrã de consola (façam Enter caso necessário):

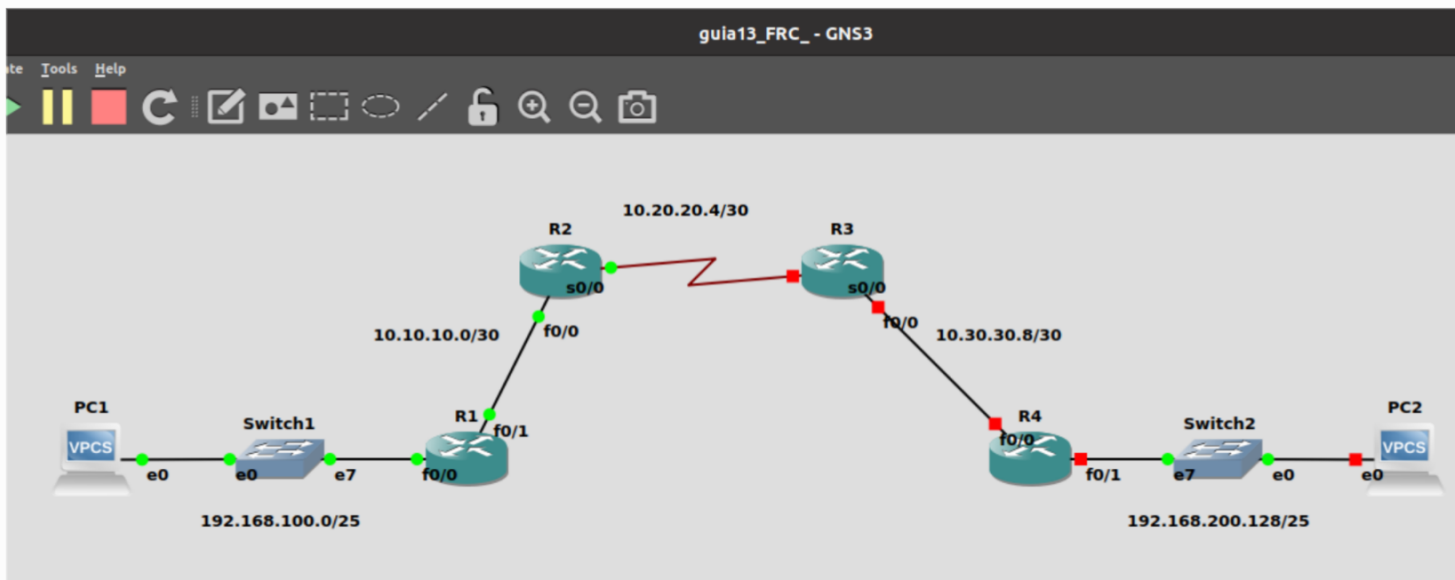


- f) O prompt **Switch>** indica que estamos ligados ao Switch mas num modo que não nos permite fazer qualquer configuração ou consulta de configuração (este modo denomina-se **User EXEC Mode**). Para passar ao modo de gestão, denominado "**Privileged EXEC Mode**", basta fazer:

Switch>en

Guião 13 – Trabalhar com os equipamentos físicos

1.



2.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#hostname R1
R1(config)#int f0/0
R1(config-if)#ip address 192.168.100.1 255.255.255.128
R1(config-if)#no shut
R1(config-if)#
*Mar 1 00:09:15.891: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:09:16.891: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#int f0/1
R1(config-if)#ip address 10.10.10.1 255.255.255.252
R1(config-if)#no shut
R1(config-if)#
*Mar 1 00:10:24.923: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:10:25.923: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R1(config-if)#do sh ip int br
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.100.1   YES manual up          up
Serial0/0      unassigned      YES unset  administratively down down
FastEthernet0/1 10.10.10.1     YES manual up          up
Serial0/1      unassigned      YES unset  administratively down down
FastEthernet1/0 unassigned      YES unset  administratively down down
FastEthernet2/0 unassigned      YES unset  administratively down down
```

```
PC1> ip 192.168.100.2/25 192.168.100.1
Checking for duplicate address...
PC1 : 192.168.100.2 255.255.255.128 gateway 192.168.100.1
```

```

R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#sh ip int br
      ^
% Invalid input detected at '^' marker.

R2(config)#do sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES unset    administratively down down
Serial0/0                unassigned      YES unset    administratively down down
FastEthernet0/1          unassigned      YES unset    administratively down down
Serial0/1                unassigned      YES unset    administratively down down
FastEthernet1/0          unassigned      YES unset    administratively down down
FastEthernet2/0          unassigned      YES unset    administratively down down
R2(config)#int f0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.252
R2(config-if)#no shut
R2(config-if)#
*Mar  1 00:18:27.807: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:18:28.807: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config-if)#int s0/0
R2(config-if)#ip address 10.20.20.5 255.255.255.252
R2(config-if)#no shut
R2(config-if)#
*Mar  1 00:19:02.171: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
*Mar  1 00:19:03.171: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R2(config-if)#do sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.10.10.2      YES manual up              up
Serial0/0                10.20.20.5      YES manual up              up
FastEthernet0/1          unassigned      YES unset    administratively down down
Serial0/1                unassigned      YES unset    administratively down down
FastEthernet1/0          unassigned      YES unset    administratively down down
FastEthernet2/0          unassigned      YES unset    administratively down down

```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#
R3(config)#
R3(config)#
R3(config)#conf t
      ^
% Invalid input detected at '^' marker.

R3(config)#int f0/0
R3(config-if)#ip address 10.30.30.9 255.255.255.252
R3(config-if)#no shut
R3(config-if)#
*Mar  1 00:00:44.983: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:00:45.983: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config-if)#int s0/0
R3(config-if)#ip address 10.20.20.6 255.255.255.252
R3(config-if)#no shut
R3(config-if)#
*Mar  1 00:01:19.827: %LINK-3-UPDOWN: Interface Serial0/0, changed state to up
*Mar  1 00:01:20.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
R3(config-if)#end
R3#sh ip
*Mar  1 00:01:24.679: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          10.30.30.9      YES manual up              up
Serial0/0                10.20.20.6      YES manual up              up
FastEthernet0/1          unassigned      YES unset    administratively down down
Serial0/1                unassigned      YES unset    administratively down down
FastEthernet1/0          unassigned      YES unset    administratively down down
FastEthernet2/0          unassigned      YES unset    administratively down down

```

```

R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int f0/1
R4(config-if)#ip address 192.168.200.129 255.255.255.128
R4(config-if)#no shut
R4(config-if)#
*Mar 1 00:06:09.347: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar 1 00:06:10.347: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
R4(config-if)#int f0/0
R4(config-if)#ip address 10.30.30.10 255.255.255.252
R4(config-if)#no shut
R4(config-if)#
*Mar 1 00:06:38.755: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:06:39.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config-if)#do sh ip int br

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.30.30.10	YES	manual	up	up
FastEthernet0/1	192.168.200.129	YES	manual	up	up
FastEthernet1/0	unassigned	YES	unset	administratively down	down
FastEthernet2/0	unassigned	YES	unset	administratively down	down

```

PC2> ip 192.168.200.130/25 192.168.200.129
Checking for duplicate address...
PC2 : 192.168.200.130 255.255.255.128 gateway 192.168.200.129

```

3.

```

R1(config)#router ospf 1
R1(config-router)#network 192.168.100.0 0.0.0.127 area 0
R1(config-router)#network 10.10
% Incomplete command.

R1(config-router)#network 10.10.10.0 0.0.0.3 area 0
R1(config-router)#exit
R1(config)#end
R1#wr
Building configuration...
[OK]

```

```

R2(config)#router ospf 1
R2(config-router)#network 10.10.10.0 0.0.0.3 area 0
R2(config-router)#networ
*Mar 1 00:31:45.295: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.100.1 on FastEthernet0/0 from LOADING to FULL, Loading Done
R2(config-router)#network 10.20.20.4 0.0.0.3 area 0
R2(config-router)#exit
R2(config)#end wr
^
% Invalid input detected at '^' marker.

R2(config)#end
R2#wr
Building configuration...
[OK]

```

```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#router ospf 1
R3(config-router)#network 10.20.20.4 0.0.0.3 area 0
R3(config-router)#network 10.20.20.4 0.0.0.3 area 0
*Mar 1 00:11:12.551: %OSPF-5-ADJCHG: Process 1, Nbr 10.20.20.5 on Serial0/0 from LOADING to FULL, Loading Done
R3(config-router)#network 10.30.30.8 0.0.0.3 area 0
R3(config-router)#exit end
^
% Invalid input detected at '^' marker.

R3(config-router)#wr
^
% Invalid input detected at '^' marker.

R3(config-router)#exit
R3(config)#end
R3#wr
Building configuration...
[OK]

```

```
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#router ospf 1
R4(config-router)#network 10.30.30.8 0.0.0.3 area 0
R4(config-router)#network 192.168.200.128 0.0.0.127 area 0
R4(config-router)#
*Mar 1 00:20:55.339: %OSPF-5-ADJCHG: Process 1, Nbr 10.30.30.9 on FastEthernet0/0 from LOADING
to FULL, Loading Done
R4(config-router)#network 192.168.200.128 0.0.0.127 area 0
R4(config-router)#exit
R4(config)#end
R4#wr
Building configuration...
[OK]
```

```
R1#
*Mar 1 00:07:15.275: %OSPF-5-ADJCHG: Process 1, Nbr 10.20.20.5 on FastEthernet0/1 from LOADING
to FULL, Loading Done
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.168.200.0/25 is subnetted, 1 subnets
O       192.168.200.128 [110/94] via 10.10.10.2, 00:01:38, FastEthernet0/1
    10.0.0.0/30 is subnetted, 3 subnets
O       10.30.30.8 [110/84] via 10.10.10.2, 00:01:38, FastEthernet0/1
C       10.10.10.0 is directly connected, FastEthernet0/1
O       10.20.20.4 [110/74] via 10.10.10.2, 00:01:38, FastEthernet0/1
    192.168.100.0/25 is subnetted, 1 subnets
C       192.168.100.0 is directly connected, FastEthernet0/0
```

```
R2(config)#
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.168.200.0/25 is subnetted, 1 subnets
O       192.168.200.128 [110/84] via 10.20.20.6, 00:01:21, Serial0/0
    10.0.0.0/30 is subnetted, 3 subnets
O       10.30.30.8 [110/74] via 10.20.20.6, 00:01:21, Serial0/0
C       10.10.10.0 is directly connected, FastEthernet0/0
C       10.20.20.4 is directly connected, Serial0/0
    192.168.100.0/25 is subnetted, 1 subnets
O       192.168.100.0 [110/20] via 10.10.10.1, 00:01:21, FastEthernet0/0
```

```
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    192.168.200.0/25 is subnetted, 1 subnets
O       192.168.200.128 [110/20] via 10.30.30.10, 00:01:31, FastEthernet0/0
    10.0.0.0/30 is subnetted, 3 subnets
C       10.30.30.8 is directly connected, FastEthernet0/0
O       10.10.10.0 [110/74] via 10.20.20.5, 00:01:31, Serial0/0
C       10.20.20.4 is directly connected, Serial0/0
    192.168.100.0/25 is subnetted, 1 subnets
O       192.168.100.0 [110/84] via 10.20.20.5, 00:01:31, Serial0/0
```



```
R4(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
192.168.200.0/25 is subnetted, 1 subnets
C      192.168.200.128 is directly connected, FastEthernet0/1
10.0.0.0/30 is subnetted, 3 subnets
C      10.30.30.8 is directly connected, FastEthernet0/0
O      10.10.10.0 [110/84] via 10.30.30.9, 00:01:51, FastEthernet0/0
O      10.20.20.4 [110/74] via 10.30.30.9, 00:01:51, FastEthernet0/0
192.168.100.0/25 is subnetted, 1 subnets
O      192.168.100.0 [110/94] via 10.30.30.9, 00:01:51, FastEthernet0/0
```

4.

```
R2(config)#service dhcp
R2(config)#ip dhcp excluded-address 192.168.100.1 192.168.100.10
R2(config)#
R2(config)#ip dhcp pool LAN100
R2(dhcp-config)#network 192.168.100.0 255.255.255.128
R2(dhcp-config)#
R2(dhcp-config)#default-router 192.168.100.1
R2(dhcp-config)#dns-server 8.8.8.8
R2(dhcp-config)#end
R2#
*Mar  1 00:43:50.899: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Building configuration...
[OK]
```

```
R4(config)#service dhcp
R4(config)#ip dhcp excluded-address 192.168.200.129 192.168.200.140
R4(config)#ip dhcp pool LAN200
R4(dhcp-config)#network 192.168.200.128 255.255.255.128
R4(dhcp-config)#default-router 192.168.200.129
R4(dhcp-config)#dns-server 8.8.8.8
R4(dhcp-config)#end
R4#
*Mar  1 00:30:34.347: %SYS-5-CONFIG_I: Configured from console by console
R4#wr
Building configuration...
[OK]
```

5.

Exemplo da configuração da autenticação para acesso à CLI e ao modo privilegiado (visto que este guião está a ser realizado em GNS3, o acesso à CLI é "direto"), apenas para o Router 1 como exemplo da configuração dessa autenticação com password não encriptada:

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#
R1(config-line)#cisco
      ^
% Invalid input detected at '^' marker.

R1(config-line)#
R1(config-line)#
R1(config-line)#
R1(config-line)#end
R1#wr
Building configuration...
[OK]
```

R1 con0 is now available

Press RETURN to get started.

User Access Verification

Password:

R1#

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logout
% Incomplete command.

R1(config-line)#end
R1#wr
Building configuration...
[OK]
R1#
*Mar  1 00:24:30.651: %SYS-5-CONFIG_I: Configured from console by console
R1#logout
```

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 15
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#
```

```
R1 con0 is now available

Press RETURN to get started.

User Access Verification

Password:
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#
```

6. Sim, continua tudo guardado nas configurações do R1 (e dos restantes routers também)

```
R1#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.100.1   YES NVRAM    up          up
Serial0/0                 unassigned      YES NVRAM    administratively down down
FastEthernet0/1          10.10.10.1      YES NVRAM    up          up
Serial0/1                 unassigned      YES NVRAM    administratively down down
FastEthernet1/0           unassigned      YES NVRAM    administratively down down
FastEthernet2/0           unassigned      YES NVRAM    administratively down down
```

```
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  192.168.200.0/25 is subnetted, 1 subnets
O       192.168.200.128 [110/94] via 10.10.10.2, 00:19:29, FastEthernet0/1
  10.0.0.0/30 is subnetted, 3 subnets
O       10.30.30.8 [110/84] via 10.10.10.2, 00:19:29, FastEthernet0/1
C       10.10.10.0 is directly connected, FastEthernet0/1
O       10.20.20.4 [110/74] via 10.10.10.2, 00:19:29, FastEthernet0/1
  192.168.100.0/25 is subnetted, 1 subnets
C       192.168.100.0 is directly connected, FastEthernet0/0
```

7. Visto que a password (usando como exemplo apenas o Router 1) está visível, isto é, não está encriptada, a forma de a ocultar é usando o comando 'enable secret [password]' para tornar a password não visível e encriptada, e, portanto, tornando a autenticação do equipamento mais segura.

```
R1(config)#enable secret cisco
R1(config)#service passord encryption
      ^
% Invalid input detected at '^' marker.

R1(config)#service password encryption
      ^
% Invalid input detected at '^' marker.

R1(config)#service password-encryption
```

R1 con0 is now available

Press RETURN to get started.

User Access Verification

Password:

R1#conf t

8. Não consegui aceder à consola de nenhum Switch no GNS3.